KUYFI

BLACK-BOX SECURITY REPORT

Before the Breach.

Contract ID	CD6QXY6STFDDPESIGCHS67SUDHXNRW3SNVIJWARKCR46QMKJZQJYYOQD
Network	TESTNET
Report ID	kyf-20261025T223848Z-4aca705d
Generated At	2026-10-25T22:38:48.381Z
Tool Version	Kuyfi 0.1.0
Schema Version	1.0.0

This report is generated by automated black-box fuzzing against a deployed Soroban smart contract. It is not a manual audit and does not constitute a formal security certification.

2. Executive Summary

Functions detected	9
UDTs detected	4
Vectors executed	39
Broadcast transactions	0
Transactions with hash	0
Total findings	0

Findings by Severity

Critical	0
High	0
Medium	0
Low	0
Info	0

3. Attack Surface

Target

Contract ID	CD6QXY6STFDDPESIGCHS67SUDHXNRW3SNVIJWARKCR46QMKJZQJYYOQD
Network	TESTNET
Bytecode size	11983 bytes

Functions Detected (9)

- `claim(recipient: Address, nullifier: BytesN<32>, proof: Proof) -> returns value`
- `close() -> returns value`
- `balance() -> returns value`
- `creator() -> returns value`
- `get_info() -> returns value`
- `next_prize() -> returns value`
- `accrue_fees() -> returns value`
- `chest_field() -> returns value`
- `__constructor(init: ChestInit) -> void`

UDTs Detected (4)

STRUCT Proof

- `a: BytesN<64>`
- `b: BytesN<128>`
- `c: BytesN<64>`

UNION DataKey

- `Info`
- `Nullifier(BytesN<32>)`

STRUCT ChestInfo

- `claims: U32`
- `closed: Bool`
- `creator: Address`
- `daily_fee_bps: U32`
- `expiry: U64`
- `fee_reserve: I128`
- `fingerprint: BytesN<32>`
- `last_fee_ts: U64`
- `prize_initial: I128`
- `prize_pool: I128`
- `prizes: Vec<I128>`
- `question: String`

- `slots: U32`
- `token: Address`
- `treasury: Address`
- `verifier: Address`

STRUCT ChestInit

- `creator: Address`
- `daily_fee_bps: U32`
- `duration_days: U32`
- `fee_reserve: I128`
- `fingerprint: BytesN<32>`
- `prizes: Vec<I128>`
- `question: String`
- `token: Address`
- `treasury: Address`
- `verifier: Address`

4. Chaos Execution

Kuyfi performs black-box fuzzing against the deployed Soroban contract using its on-chain contract specification.

Vectors executed	39
Broadcast transactions	0
Transactions with hash	0
Verification transactions	0

5. Findings

NO REPORTABLE FINDINGS

No reportable security findings were produced by this audit run. This does not constitute a formal proof that the contract is vulnerability-free.

6. Verification

Real Testnet transactions broadcast during this audit run, provided so every claim above can be independently verified on-chain.

No transaction ever reached broadcast during this run.